

SELF-SUPERVISED PRETRAINING · 3D SEGMENTATION

MISFIT: A Self-Supervised Pretraining Toolkit for 3D Medical Imaging

A Preliminary Test Across Three BraTS 2026 Segmentation Tasks

Adrian Celaya^{1,2}, Yogesh Kumar¹, Awj Twam¹, Beatrice Riviere², David Fuentes¹, Tucker Netherton¹

¹The University of Texas MD Anderson Cancer Center, Houston TX ²Rice University, Houston TX

Brain Metastases Task 1 · n = 1,196	Pediatric Tumors Task 2 · n = 294	BraTS-GoAT Task 3 · n = 1,351
---	---	---

Introduction

TODO – Motivation for MIST/MISFIT, the three BraTS 2026 tasks (metastases, pediatric tumors, BraTS-GoAT), and the two questions this study asks (system-level deployment vs. isolated pretraining effect).

Methods

TODO – Data (Table 1: task, cohort, training/held-out counts, regions), preprocessing, MISFIT self-supervised MAE pretraining, the three network arms (pocket nnUNet baseline, MISFIT-pretrained SwinUNETR-V2, from-scratch SwinUNETR-V2), and the evaluation protocol (cross-validation vs. held-out, BraTS-style rank, Wilcoxon test).

References

TODO – Reference list.

Results

TODO – Table 2 (average rank + p-values), Table 3 (per-region metrics), Figure 1 (validation loss curves), Figure 2 (qualitative comparison).

Discussion & Conclusion

TODO – Held-out validation does not favor the pooled MISFIT pipeline over the pocket nnUNet baseline; matched-control comparison shows MAE pretraining helps when architecture/patch size/budget are fixed; final challenge submission uses the compliant baseline.